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## THE HISTORY OF THE DECLINE OF THE LARGER MAMMALS OF THE LAKE CHILWA BASIN

C. O. Dudley

Before the establishment of agricultural settlements, the vegetation of the Lake Chilwa catchment was as varied as any area in Malaŵi. Evergreen forests and grasslands covered the Zomba and Mulanje Mountains. *Brachystegia* woodland extended from these Mountains' lower slopes throughout the Plateau. Below 750 metres, *Terminalia* swamp woodland and lowland plain woodland spread across the Chilwa and Phalombe Plains eventually reaching the floodplains and swamps that surround Lake Chilwa. Jackson (1972), Howard-Williams (1973, 1977) and Howard-Williams and Walker (1974) have described and mapped what are thought to be the "natural" plant communities of this area. Within this ecosystem the larger mammals must have been as diverse and numerous as anywhere in Central Africa.<sup>(1)</sup>

In those days, man and animal, on the whole, lived in "equilibrium". One of the factors contributing to this equilibrium was the practice of totemism, a common custom in societies wherever man is largely dependant on the hunting of wild animals (Leach, 1967). Totemism meant that each clan had a special relationship to a particular animal. These animals were protected since, according to tradition, a man was not allowed to kill either the animals of his or his neighbour's clan (Dr. M. Schoffeleers as cited in Hayes, 1972). The Yao may once have been elephant totemic, but altered their customs as a result of the 18th century boom in the ivory trade (Vaughan, 1978). They began killing elephants but an elaborate cleansing ritual was required. Also, the fact that the ivory had to be channeled through the chief ensured that he could control the number of elephants killed in his area. Chiefly control of the fishing season in Lake Chilwa is mentioned in oral testimonies (Vaughan, 1978) and may show an awareness of the importance of conservation. The breakdown of this *de facto* indigenous conservation technique may have been accelerated by European laws which took control of game away from the Chiefs (see below).

Today the Lake Chilwa Basin, particularly at elevations below 1,000 metres, is greatly changed (Fig. 1). Only relic patches of the

(1) Because of its inaccessibility and the dearth of historical records, most of the discussion included in this paper refers to the western, northern and southern sides of the Lake only. The eastern side of the Lake Basin is unknown to the author but probably has undergone a similar historical change in its large mammal fauna.

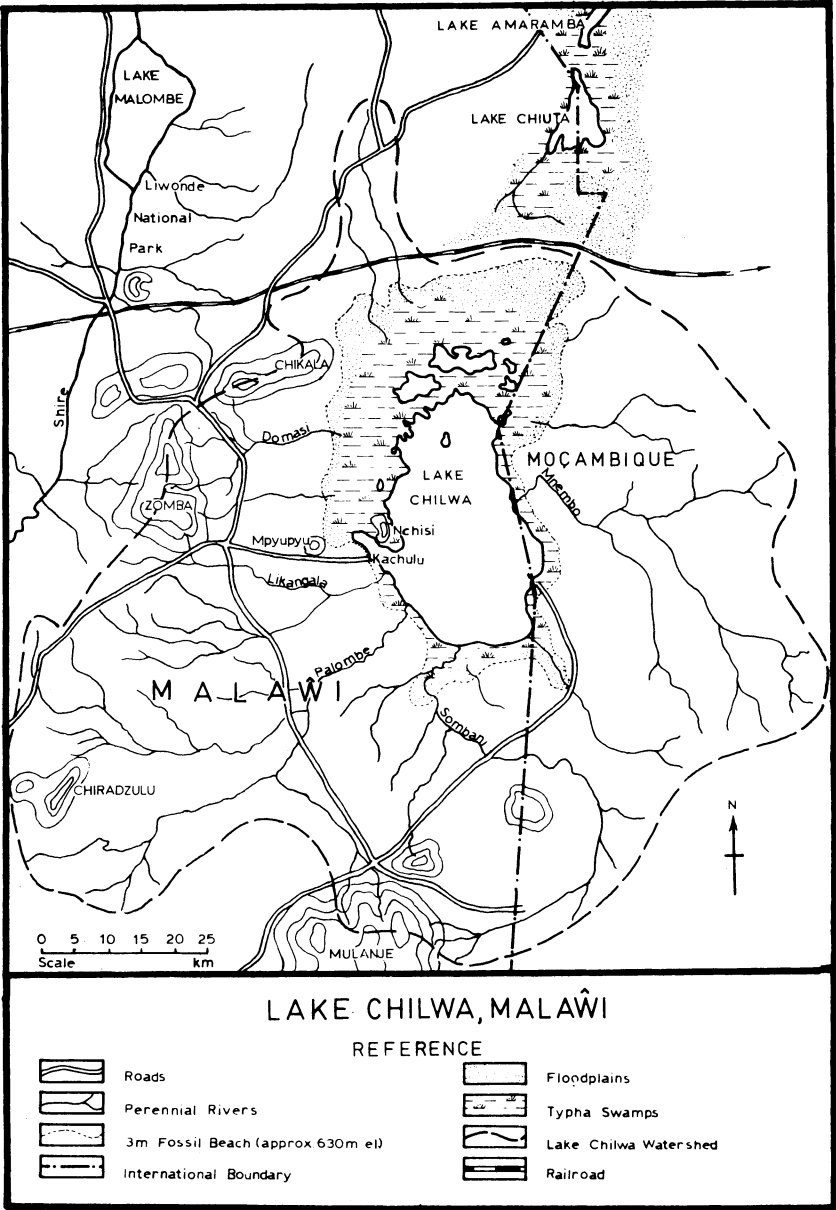


Fig. 1 A map of Lake Chilwa Basin and its environs. Only that part of the Lake's catchment lying below 1,000 m (90%) will be considered in this paper.

woodlands remain and agricultural activities dominate the landscape. Lake fishing communities are established on the Lake's shores and islands. Much of this change occurred during the last 120 years and it is the purpose of this paper to document these changes via historical accounts and recent observations.

It is incorrect to analyze the decline of the mammal populations only in terms of the swamps and flood plains. (1) Most historical writers talked of the Chilwa Plains in a vague, undefined way, undoubtedly including a considerable portion of the woodlands above 700 metres. (2) Larger mammals are so mobile that they are seldom tied to specific bodies of water. Even hippos move as much as 12 kilometres from permanent water, remaining in temporary pools until the onset of the dry season. (3) Populations of many large mammals require a large heterogeneous physical area to support them. Because of historical vagueness, mammal mobility and extensive home ranges, all the Basin below 1,000 metres will be included in this analysis<sup>(2)</sup>. Table 1 lists the larger mammals known to have occurred or thought to have occurred in the Lake Chilwa Basin (below 1,000 metres) in the 19th century.

**Table 1:** Large mammals known to have occurred or thought to have occurred in the Lake Chilwa Basin below 1,000 metres in the 19th century. Those species marked with an asterisk (\*) are inferred as they were not recorded historically but currently exist in small numbers. The remaining species were reported by various historical sources.

|                      |                                           |
|----------------------|-------------------------------------------|
| Yellow baboon*       | <i>Papio cynocephalus</i> (L)             |
| Vervet monkey*       | <i>Cercopithecus aethiops</i> (L)         |
| Cape clawless otter* | <i>Anonyx capensis</i> (Schniz)           |
| Civet-cat*           | <i>Viverra civetta</i> Schreber           |
| Water mongoose*      | <i>Atilax paludinosus</i> (Cuvier)        |
| Spotted hyaena       | <i>Crocuta crocuta</i> (Erleben)          |
| Lion                 | <i>Panthera leo</i> (L)                   |
| Leopard              | <i>P. pardus</i> (L)                      |
| Ground pangolin*     | <i>Manis temmincki</i> Smuts              |
| African elephant     | <i>Loxodonta africana</i> (Blumenbach)    |
| Bushpig              | <i>Potamochoerus porcus</i> (L)           |
| Warthog              | <i>Phacochoerus aethiopicus</i> (Pallas)  |
| Hippo                | <i>Hippopotamus amphibius</i> (L)         |
| Grey duiker          | <i>Sylvicapra grimmia</i> (L)             |
| Oribi                | <i>Ourebia ourebi</i> (Zimmermann)        |
| Klipspringer         | <i>Oreotragus oreotragus</i> (Zimmermann) |
| Reedbuck             | <i>Redunca arundinum</i> (Boddaert)       |
| Waterbuck            | <i>Kobus ellipsiprymnus</i> (Ogilby)      |
| Impala               | <i>Aepyceros melampus</i> (Lichtenstein)  |

(2) Ninety percent of the Lake Chilwa Basin (catchment) lies below 1000m.

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|                           |                                                |
|---------------------------|------------------------------------------------|
| Sable                     | <i>Hippotragus niger</i> (Harris)              |
| Lichtenstein's hartebeest | <i>Alcelaphus lichtensteini</i> (Peters)       |
| Eland                     | <i>Taurotragus oryx</i> (Pallas)               |
| Bushbuck                  | <i>Tragelaphus scriptus</i> (Pallas)           |
| Greater kudu              | <i>T. Strepsiceros</i> (Pallas)                |
| Nyasa brindled gnu        | <i>Connochaetes taurinus johnstoni</i> Sclater |
| African buffalo           | <i>Syncerus caffer</i> (Sparrman)              |
| Rhino                     | <i>Diceros bicornis</i> (L)                    |
| Burchell's zebra          | <i>Equus burchelli</i> (Gray)                  |

## THE ENVIRONMENT

If the historical decline of the larger mammals of the Lake Chilwa region is to be understood, the interrelated environmental factors of Lake, vegetation and people must be examined. However, this examination is fraught with difficulties as many of the early observations abound with exaggeration and contradictions.

**The Lake:** Livingstone was the first European to describe Lake Chilwa, first seeing the Lake on April 18, 1859. (Wallis, 1956). In his dispatches to England he greatly exaggerated its size, claiming it to reach Palombe Mountain. His companion Kirk mapped it only slightly smaller, but his description of the extent of dry land between the Lake Swamp and the mountains of Chikala and Mpyupyu indicated a shoreline little different from that of today (Foskett, 1965). Stewart's map, from observations made in 1862, was nearly identical (Wallis, 1952). O'Neill (1884), after exploring much of the Lake's eastern and northern shore, described and mapped the Lake more realistically. His map showed the Lake similar in extent to its maximum size today, being below the fossil three metre beach terrace. Although Johnston's (1897) published map agreed with O'Neill's, Johnston still overestimated the Lake's size in the text of his book.

The Lake is endorheic and, thus, is inherently unstable being reported "dry" in 1879 (Buchanan, 1885; MacDonald, 1882 O'Neill, 1884) 1900 (Duff, 1903), 1922 (Dixey, *et al.* 1955), 1934 (Garson and Campbell - Smith, 1958), 1968 (Kalk, 1970) and 1973. The historical contradictions referred to in the preceding paragraph have been attributed to this instability. However, Livingstone's estimate would have required a lake over 30 metres higher than present, emptying into Lake Chiuta! This vast amount of water would have had to evaporate almost completely by 1879 to produce the dry lake

O'Neill, MacDonald and Buchanan recorded. Even Garson's (1960) estimate of recent lake depths 9 metres above current levels seems unreasonable considering the historical records of the dry phases. The evidence suggests, then, that historical fluctuations in the Lake's level have occurred only within the three metre fossil beach contour.

Fluctuations in Lake level within this contour would only have caused short period changes in large mammal populations of the Basin. During times of drought, open water, if it existed, could only have been reached by moving along the perennial rivers entering the west side of the Lake. The narrow band of flood plain grasslands would have been quickly lost through over grazing. To escape these adverse conditions the mammals need only to have emigrated to better habitats north and east of the Lake (e.g. Upper Shire Valley, Lake Chiuta and Lake Amaramba Basins), returning when conditions improved.

**The Vegetation:** Early explorers such as Livingstone, Kirk, and Stewart described the Shire Highlands and the Chilwa Basin as "well wooded". Yet, Kirk's account in 1859 stated that the woodland was open and, below about 700 metres, dominated by *Upaca* sp. This tree often indicates woodland that has been degraded by past fire and cultivation. Later, O'Neill (1884) presented a more variable picture. The northeast side of the Lake, above the swamp edge, was described as grassland with widely scattered scrub and trees while the vegetation of the north, including the sand bar, was densely wooded. Emphasizing this difference is the fact that Kirk's description was made in April when the vegetation was its thickest, while O'Neill's observations took place in October when fire would have been expected to have taken its toll of the herbaceous vegetation. The following year Buchanan (1885) painted the same picture for the Highlands around Zomba "well wooded".

Duff (1903) was the first to reveal less than a virgin woodland cover. Although he described the plains as being forested he considered the woodland to be of poor quality due to shifting cultivation. The Nyasaland Handbook of 1908 (Murray, 1908) and Lyell (1912) again reverted to the "well wooded" depiction. But, by 1922 and 1932, the Handbook gave a less uniform analysis. Immense tree-less plains were said to occur between the Lake and the Palombe and Sombani Rivers with only small patches of park-like forests near Sombani.

Today, below 1,000 metres, only a few small stands (less than 5 hectares each) of *Brachystegia* and *Upaca* woodland can be found on the Chilwa Plains. A larger unbroken section (approximately 10,000

hectares) of lowland plains woodland still exists on the Palombe Plain. This woodland abuts onto the swamp near the mouth of the Sombani River. The remaining land is cultivated or cultivation fallow (secondary grassland) or floodplain grassland. The river dambos and floodplains probably represent the only areas of “original” grassland. This relatively restricted open grazing meant that browsing (e.g. kudu, bushbuck, rhino) and mixed feeding (e.g. elephant, eland, impala) herbivores were probably dominant before the advent of intensive cultivation. However, the deforestation of the early 1800’s and the subsequent human depopulation of the region (see below) may have shifted the balance in favour of the grazers (e.g. gnu, zebra, sable) during the last half of the 19th century.

**The People:** According to Vaughan (1978), it is generally believed that the southern half of Malaŵi has been densely populated for a long time, at least by African standards. From historical records it is difficult to determine just how populated the Lake Chilwa Basin was. The slave trade, colonial exploitation and settlement, drought and disease all influenced population density and dispersion.

Livingstone, Kirk and Stewart all reported a number of apparently substantial villages along their routes through the Highlands and to Lake Chilwa. Rowley (1881) noted that in 1861 the Highlands were heavily populated. In 1879 MacDonald (1882) described the inhabitants at the base of Zomba Mountain as numerous and quotes Chief Malemia as speaking of “the great number of people that could be reached (evangelized) from Chirwa”. Yet, later writers remarked on the absence of people on the surrounding Lake plains. It is Vaughan’s (1978) thesis that the intensification of the slave trade and inter-tribal warfare during the period of 1865 - 1890 caused the abandonment of much of the lowlying alluvial land in southern Malaŵi. This is supported in part by Buchanan (1865), MacDonald (1882), Last (1887), Hetherwick (1887), and Drummond (1888). MacDonald wrote that “the people who lived (near) Cherasulu (Chiradzulu) were all full of terror and had fled up the mountain to spots... inaccessible”. He also found heavily overgrown village sites abandoned from the time of the N’goni wars. Buchanan stated that 2,000 Nyanja were concentrated on Nchisi Island to escape slavers. By 1887 Hetherwick estimated the number to be 3,000. He also reported that the Yao had taken to the hills after the N’goni invasion of 1883. Last claimed that the fertile spurs of Zomba Mountain had been abandoned as the result of raids. Oral testimonies have revealed that many Nyanja villages moved up onto the top of Zomba Mountain and remained there for at least one generation (Vaughan,

1978). Drummond described the western Chilwa plains as a major slave route. The plains were underpopulated and the people of the Lake edge were scattered and poor.

This “distorted distribution of people may have contributed to a temporary increase in game in the Chilwa Basin (Watson, 1973). Vaughan (1978) contends that the colonial administration maintained this population dispersion for the benefit of settlers and hunters during the early part of the 20th century. Gun control is put forth as an example. Prior to 1900 most African families in the area had guns but by 1902 the number was only 205 as few Africans could afford the license fee (Central African Times 9/14/1902; Duff, 1903). Game exploitation was increasingly a white man’s prerogative; control had been taken from the Chiefs.

In succeeding years, population pressure, coupled with political activity, broke down this situation and the western plains of the Lake became increasingly populated. By 1966 this area had some of the highest rural densities in Southern Africa (Agnew, 1970) and the 1976 census (Anonymous, 1978) has shown a further large average per annum increase in population of over 2.5 percent in the Traditional Authorities surrounding the Lake. The highest rate of increases occurred in Chief Mkumbira’s area, the T.A. which encompasses most of the Lake and its Islands. In 1928 Dixey reported that Nchisi Island no longer was densely populated, having no more than 100 people. By 1966, before the Lake dried up, Nchisi again had 2,000 inhabitants, principally involved in fishing. The 1976 census listed Nchisi with 1848 people, suggesting that the island had reached its limit under the ideal conditions of a full lake.

### **The Mammals**

Livinstone’s and Kirk’s explorations of the Chilwa Basin revealed little game. Hippo were considered numerous in the Lake and although Kirk reported seeing much buffalo spoor, no other game was seen.

Rawley (1881) claimed that the central part of the Shire Highlands were without “wild animals” in 1861. O’Neill (1884) mentions game as being numerous just north of the Lake but saves his enthusiastic descriptions for the lands beyond the sand bar and west of the Lakes Chiuta and Amaramba. In 1888, Drummond painted a slightly improved picture with zebra abounding on the western shore and antelope (unnamed) plentiful. Johnston’s (1897) descriptions were more specific and possibly suggested an increase in game in the



area. Just how much this reflected the “distortions” in population dispersions of the people during the period of 1870 - 1880, referred to in Vaughan ((1978) and Watson (1973), is unknown. Elephant and eland were occasionally seen in Zomba Township and hunting dogs were known on the Mountain. Hippo were so numerous in the waterways that Johnston (1894) advocated their “merciless extermination”. Maughan (1929) in 1894 and Lyell much later in 1923, substantiated the great diversity of the larger mammals found within the Lake Basin. They recorded impala, eland, bushbuck, duiker, reedbuck, sable, hartebeest, waterbuck, kudu, gnu, klipspringer, oribi, buffalo, zebra, rhino, elephant, hippo, bushpig, warthog, hyaena, leopard, and lion. The sizes of these populations are unknown.

When the British Colonial Administration was established in 1891, the *status quo* of animals and man was drastically altered. The Lake's plains were but 30 kilometres from the administrative capital of Zomba and the farms of the Highlands. The custom among officials, settlers, and visitors was to spend weekends hunting game on the Chilwa Plains with the best fire arms that could be imported. (Maughan, 1929).

The rising impact of the European on the big game of the region is reflected in the official ivory export figures during the last decade of the 19th century. By 1893, ivory exports claimed 83 percent of the export trade (Baker, 1961). This represented approximately 19,316 kilos or the equivalent of 500 - 1,000 elephants! Johnston (1897) estimated that almost as much ivory was being exported illegally. By 1899 elephant were becoming scarce and only 2,454 kilos were exported. In spite of this drop, 72,727 kilos, representing 2,500 -3,000 elephant, were officially exported from 1891 to 1904. During the next sixty years the export averaged about 4,545 kilos per year. Similar export trends, though on a smaller scale, were recorded for hippo teeth and rhino horn.

An incident of interest during this period was that of the Rinderpest outbreak that, it is claimed, by-passed Nyasaland (Johnston, 1897). After examining the historical records Hobbs (1976) concluded that this “epidemic had little direct effect on Nyasaland's faunal communities.”

The beginning of the general decline in some species of game was noted as early as 1897. “The gnu seems to diminish rapidly before the European and Native sportsman and in places where he was once abundant is no longer met with”. (Johnston, 1897). And Johnston's

earlier admonition in (1894) to exterminate the hippo was proving successful. They were no longer a source of danger to navigation (Vaughan-Kirby, 1899; Webb, 1897) and were only seen occasionally in Lake Chilwa (Johnston, 1897). In response to these trends, the Colonial Administration established the Lake Chilwa Game Reserve in 1897. This action set aside a large area on the Palombe Plains south of Likangala River and adjacent to the southwest Lake shore. The Reserve was very unpopular with the African agriculturists of the Plains and, after strong protests against it, it was withdrawn in 1902. The Lake Chilwa Reserve might have been more successful if it had been smaller and administered adequately.

In 1908 Handbook of Nyasaland (Murray, 1908) still reported that game was extremely plentiful within the District (with the exception of the elephant) and professional hunters, such as Lyell (1912) were recommending the Colony as an ideal place for the Big Game Hunters (Fig. 2). But, by 1922, the game populations were deteriorating throughout the Southern Region (Murray, 1922). No hippo were recorded from the Lake and only limited numbers of kudu, sable, reedbuck and hartebeest were still to be found in the forested hills of Chikala and the woodlands near Sombani. According to G. D. Hayes (pers. comm.), "One might have thought the Chikala Hills to be fairly inaccessible in those days, but the old Zomba-Malakotera military road passed through and with guns handy and easy targets available, the mammals suffered". Tsetse, after being prominent in 19th century reports (Duff, 1903; Kirk, in Foskell, 1966; MacDonald, 1882) seemed no longer a threat by

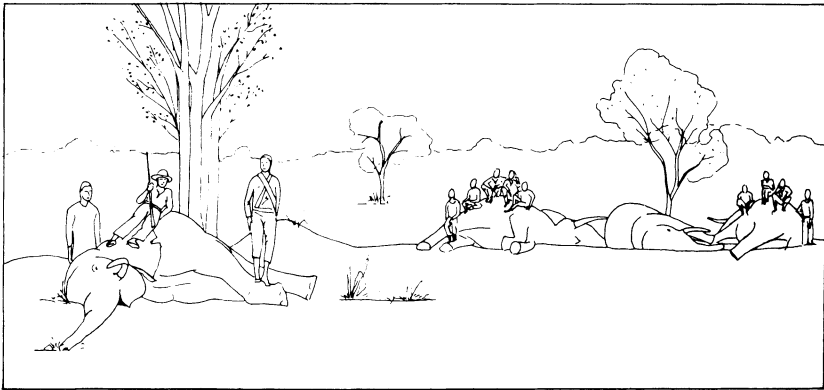


Fig. 2 The white hunter with his trophies. Sketch from a photograph in Sutherland (1912).

1922. The vegetation and the mammalian hosts necessary for their survival were gone.

The first mammal known to become extinct in Malaŵi was the Nyasa brindled gnu, (Fig. 3) an antelope limited to the short grass plains around the perimeter of the Lake. A few individuals were reported present in 1922 (Murray, 1922) but by 1928 this antelope no longer occurred in Malaŵi (Sweeney, 1959). The enthusiastic descriptions of earlier hunters were now greatly moderated “quality . . . rather than quantity should be the aim of the hunter”, (Wood, 1932).

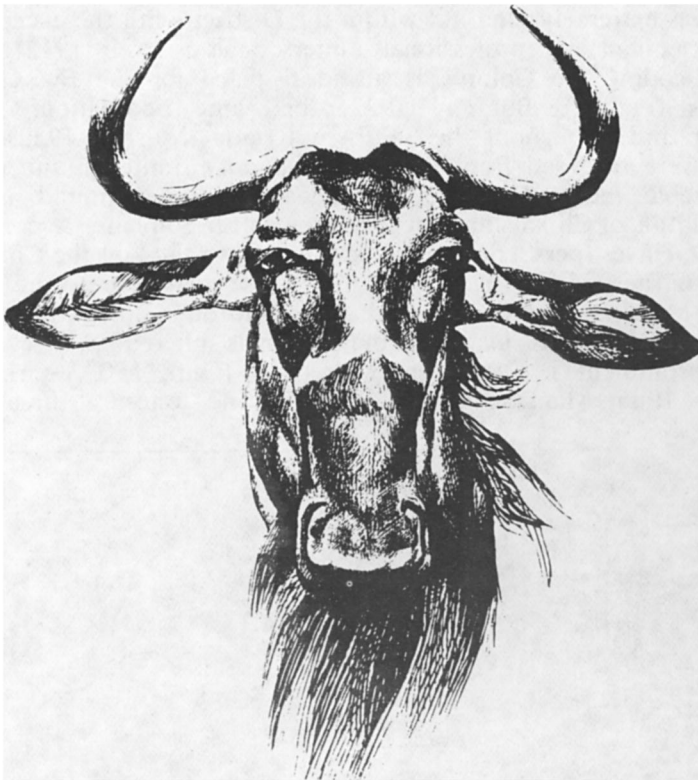


Fig. 3 The Nyala brindled gnu, *Connochates taurinus johnstoni*, an early 20th century casualty to colonial hunting. From an early sketch by H. H. Johnston (1897).

In the 1930's game was scarce in the Chikala Hills and only warthog, bushpig, baboon, and bushbuck and the occasional lion or leopard were seen on the western slopes of the Basin (Murray, 1932). The southern end of the Basin, except for water birds, "was a dreary stretch of country without game of any kind". Garson (1960) paints a similar picture and Sweeney (1965), after several extensive flights over the flood plains of the north and south could report seeing only 15 reedbuck, 1 bushbuck and several duiker. The situation is little changed today. Only those mammals capable of maintaining their populations in the vicinity of man remain. These include the vervet monkey, baboon, bushbuck, duiker, civet cat, mongoose and hyaena. The latter still take the occasional cow in the north (Hobbs, 1976). A few additional species of the larger mammals may still live in very low numbers in the Forest Reserves on the hills surrounding the Plains.

The wildlife of the Chilwa Basin has not been entirely lost to southern Malawi. Early pressure was put on the Colonial Government for a more active conservation policy by the National Fauna Preservation Society (an extra-governmental organization founded in 1947 by concerned citizen conservationists). Its submissions contributed to the establishment of Majete and Mwabvi game Reserves (1955) and Nyika, Malawi's first National Park (1966) and to the continued existence of the nyala antelope in Lengwe National Park (Hayes, 1973). After Independence, Malawi pursued a vigorous conservation policy eventually establishing and maintaining four game reserves and four national parks.

The latest of these is Liwonde National Park. Declared in 1972, it includes 40 kilometres of the Shire river and Lake Malombe shore line and contains approximately 160,000 hectares of the east side of the Upper Shire Valley. This once excellent game country is only 40 kilometres northwest of Lake Chilwa beyond the Chikala Hills. Today, thanks to protection, this Park is populated by most of those species which 19th century explorers and hunters reported to be present in the Lake Basin (Dudley and Stead, 1976; Stead and Dudley, 1977). Future introductions into Liwonde will enhance this likeness.

Cattle were first introduced into this area in 1880 (MacDonald, 1882) and now, instead of the mixed herds of game, only cattle find pasture within the Basin. A few large, intensely managed farms (e.g. Rathdrum, Chimpeni) maintain quality beef and dairy herds on the higher elevations (750 metres) but the vast majority of cattle (27,000

head in Lake Chilwa Basin (Schmidt, 1969) are found on the flood plains and secondary grasslands immediately surrounding the Lake. The grasslands have probably suffered as a result of this ecological simplification.

Grazing by a mixed herbivore community favours higher levels of nutrients and productivity (Bell, 1969, Vesey-FitzGerald, 1969). By means of a successional grazing pattern, wild herbivores develop a mosaic of grass species. Cattle tend to break down this successional pattern and thereby reduce total productivity and, ultimately, may cause the loss of palatable species. Vesey-FitzGerald (1970) the late well known grassland ecologist, noted that the Lake Chilwa grasslands were showing signs of overgrazing. This was probably a result of the herding behaviour and catholic diet of cattle.

The richness of the flood plains and Lake also may have suffered from the loss of nutrients when the droppings of the original game no longer were available to the Chilwa system. Dung washed into the Lake in flood adds significantly to its nutrient content. McLachlan (1971) attributed a release of nutrients from the Lake Kariba shoreline following rains to a release from dry grass and dung. The dung from the present large herds of cattle may correct this situation.

Intensive cattle husbandry will undoubtedly improve the production of protein on the Lake Chilwa Plains but it is unlikely to reach levels of productivity equal to that of the "original" ecosystem.

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